

2.1

a)

	PCA dimension reduction	Spectral embedding
optimization goal	maximize the variance of the data projected onto k orthogonal directions	minimize $\sum_{i,j} w_{ij} \ y_i - y_j\ ^2 = 2 \text{tr}(Y^T L Y)$ subject to $Y^T Y = I$
matrix for the eigendecomposition	covariance matrix $C = \frac{1}{n-1} D^T D$ of the mean-centred D	Laplacian $L = \Lambda - W$ of the similarity graph
size of the matrix	$d \times d$	$n \times n$
k best eigenvalues	$\lambda_{\max-k+1}, \dots, \lambda_{\max}$ (largest)	$\lambda_2, \dots, \lambda_{k+1}$ (smallest except $\lambda_1 = 0$)
new data after transformation	DP_k with $P_k = [v_{\max-k+1}, \dots, v_{\max}]$ ($n \times k$)	$[v_2, \dots, v_{k+1}]$ ($n \times k$)

b) $\sqrt{53} = d((0,0), (7,-2))$ is the largest distance between two points, so the similarities lie in $[0, 1]$. With the points numbered 1–6 in the given order, the 2-nearest neighbour graph has the edges 1–2, 2–3, 4–5 and 5–6 with similarity 0.717, 1–4 with 0.566, 2–4 with 0.752, 3–4 with 0.693 and 4–6 with 0.434.

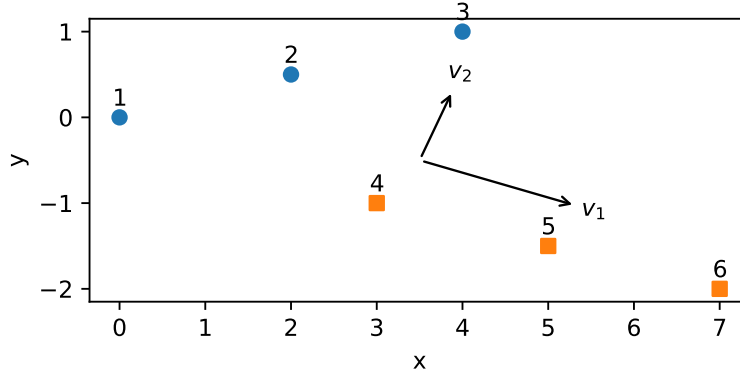
$$L = \begin{pmatrix} 1.282 & -0.717 & 0 & -0.566 & 0 & 0 \\ -0.717 & 2.186 & -0.717 & -0.752 & 0 & 0 \\ 0 & -0.717 & 1.410 & -0.693 & 0 & 0 \\ -0.566 & -0.752 & -0.693 & 3.161 & -0.717 & -0.434 \\ 0 & 0 & 0 & -0.717 & 1.434 & -0.717 \\ 0 & 0 & 0 & -0.434 & -0.717 & 1.150 \end{pmatrix}$$

has the eigenvalues 0, 0.598, 1.342, 1.993, 2.851, 3.839 with the eigenvectors in the columns of

$$V = \begin{pmatrix} 0.408 & 0.390 & 0.711 & -0.044 & 0.399 & 0.120 \\ 0.408 & 0.344 & -0.027 & -0.039 & -0.796 & 0.282 \\ 0.408 & 0.334 & -0.700 & -0.076 & 0.443 & 0.173 \\ 0.408 & 0.036 & -0.041 & 0.105 & -0.098 & -0.900 \\ 0.408 & -0.493 & 0.013 & 0.727 & 0.047 & 0.244 \\ 0.408 & -0.611 & 0.043 & -0.672 & 0.005 & 0.080 \end{pmatrix}.$$

The 1-dimensional presentation is $v_2 = (0.390, 0.344, 0.334, 0.036, -0.493, -0.611)$. It preserves the clustering structure: points 1–3 lie together at 0.33–0.39 and points 4–6 below them, although point 4 lies between the clusters.

c) The PCA presentation orders the points 1, 2, 4, 3, 5, 6, so point 4 falls between points 2 and 3 and the clusters overlap, whereas the spectral embedding keeps them apart. PCA keeps the direction of largest variance and ignores which points are neighbours. Here v_1 runs along both elongated clusters, while the clusters are separated in a direction of small variance that is discarded.



2.2

1.

Run	Iteration	Label of point								Representative	
		1	2	3	4	8	10	11	100	c_1	c_2
K-Means, (8), (11)	1	1	1	1	1	1	2	2	2	3.6	40.33
	2	1	1	1	1	1	1	1	2	5.57	100
	3	1	1	1	1	1	1	1	2	5.57	100
K-Means, (1), (8)	1	1	1	1	1	2	2	2	2	2.5	32.25
	2	1	1	1	1	1	1	1	2	5.57	100
	3	1	1	1	1	1	1	1	2	5.57	100
K-Medians, (8), (11)	1	1	1	1	1	1	2	2	2	3	11
	2	1	1	1	1	2	2	2	2	2.5	10.5
	3	1	1	1	1	2	2	2	2	2.5	10.5
K-Medians, (1), (8)	1	1	1	1	1	2	2	2	2	2.5	10.5
	2	1	1	1	1	2	2	2	2	2.5	10.5
K-Medians, (1), (100)	1	1	1	1	1	1	1	1	2	4	100
	2	1	1	1	1	1	1	1	2	4	100

K-Means ends with $\{1, \dots, 11\}$ and $\{100\}$ from both initializations, because the outlier pulls the mean of its cluster away. K-Medians is more robust to the outlier and finds $\{1, 2, 3, 4\}$ and $\{8, 10, 11, 100\}$, unless the outlier is an initial representative, in which case it stays a singleton cluster as with K-Means. The result thus depends on the algorithm and the initialization, and an outlier can take a whole cluster.

2.

x	#1			#2		
	a	b	$S(x)$	a	b	$S(x)$
1	2.00	31.25	0.936	5.33	99.00	0.946
2	1.33	30.25	0.956	4.50	98.00	0.954
3	1.33	29.25	0.954	4.00	97.00	0.959
4	2.00	28.25	0.929	3.83	96.00	0.960
8	32.33	5.50	-0.830	4.50	92.00	0.951
10	31.00	7.50	-0.758	5.50	90.00	0.939
11	31.00	8.50	-0.726	6.33	89.00	0.929
100	90.33	97.50	0.074			0
SC			0.192			0.830

No. SC prefers #2, but the outlier is so far away that it dominates the mean distances. In #2 all other points get $S \approx 0.95$ although the groups $\{1, 2, 3, 4\}$ and $\{8, 10, 11\}$ are merged, and in #1 the points 8, 10 and 11 get negative silhouettes although they form a natural group.

2.3

- a) Clusters and classes coincide, so $I(C, D) = H(C) = H(D) = 1$ bit and $NMI = 1$.
- b) Each cluster contains one point of each class, so $P(C_i, D_j) = \frac{1}{4} = P(C_i)P(D_j)$, $I(C, D) = 0$ and $NMI = 0$.
- c) $H(C) = 2$ bits and $H(D) = 1$ bit, and every singleton cluster is pure, so $I(C, D) = H(D) = 1$ bit and $NMI = 1/\sqrt{2} \approx 0.71$. This is not intuitive, because the classes are irrelevant to the clustering.
- d) Independence holds for the true probabilities, but NMI is computed from relative frequencies in the data. With singleton clusters, $P(C_i, D_j) = 1/n$ for the class of the point and 0 otherwise, which never equals $P(C_i)P(D_j) = P(D_j)/n$. The estimated mutual information is therefore always $H(D) > 0$, and this bias grows with the number of clusters.